

Show understanding of the need for and examples of the use of compression

9618_s25_qp_11_sc_2.b.i

QUESTION

- (b) (i) Explain the reasons why a video is compressed before it is transmitted using real-time bit streaming.

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..... [4]

MARK SCHEME

2(b)(i)	1 mark each to max 4 e.g. • Video is data-intensive • The file size needs reducing in order to • reduce the amount of bandwidth used • and reduce buffering • This means people are not behind in the conversation • and people with lower bandwidth can still take part	4
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QUESTION

(b) The student compresses the file before it is emailed to their teacher as an attachment.

(i) Explain the benefits to the teacher of the attachment being a compressed file.

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MARK SCHEME

4(b)(i)	1 mark for each bullet point (max 3) e.g. • Less of the teacher's storage space is used • ... so more student work can be stored • Transmission time is reduced • ... so the teacher does not have to wait as long for it to arrive • Bandwidth usage is reduced • ... so other transmissions are not adversely affected • Less data allowance is used on the teacher's email system	3
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QUESTION

(iii) The bitmap video is **not** compressed before transmission to the VR headset.

Give **two** reasons why the video does **not** need to be compressed.

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2

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MARK SCHEME

2(d)(iii)	1 mark each to max 2 : • Dedicated connection to the headset // not sharing bandwidth • Already fast connection that can transmit the data without slowing • Video may already be a small file size and does not need further reduction • Video is not saved so storage is not an issue in the headset	2
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9618_w23_qp_13_sc_5.b.i

QUESTION

(b) The program is distributed by downloading the source code and its library files from a web server.

(i) Explain the reasons for compressing the files.

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..... [2]

MARK SCHEME

5(b)(i)	1 mark for each bullet point (max 2). • To reduce the time it takes to download the program files from the web server // to upload them to the server in the first place • To reduce the amount of storage space used on the web server // user's device	2
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9618_w23_qp_11_sc_1.c

QUESTION

(c) Explain why a bitmap image is often compressed before it is attached to an email.

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MARK SCHEME

1(c)	1 mark for each bullet point (max 2) • Reduced bandwidth usage when transmitting the message • Reduced transmission time from email client to email server • Reduced storage space on the email • Email accounts often have a maximum size for an attachment	2
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9618_s23_qp_11_sc_1.c.i

QUESTION

(c) The photographs are compressed before they are uploaded to a web server. Customers download the photographs from this web server.

(i) Explain the reasons why compressing the photographs will benefit the customers.

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MARK SCHEME

1(c)(i)	1 mark each Examples: • The customers will be able to download the photographs in less time • ...and they will take less of the customer's bandwidth • The photographs will take up less space on the customer's storage medium • ...therefore the customers can store more images • ...and will have more space for other files	3
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QUESTION

(b) Bobby wants to email the sound file to his school email address. He compresses the file before sending the email.

(i) Explain the reasons why Bobby compresses the sound file.

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MARK SCHEME

7(b)(i)	1 mark per bullet point • Reduces the file size • Faster to transmit/download • Original file is too large for email storage/attachment	2
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Show understanding of lossy and lossless compression and justify the use of a method in a given situation

9618_w25_qp_12_sc_6.a

QUESTION

- 6 (a) A sound file is compressed by reducing the sampling rate.

State whether this is lossless or lossy compression. Justify your choice.

Type of compression

Justification

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[1]

MARK SCHEME

6(a)	1 mark for correct answer Type of compression: lossy Justification: There will be fewer samples per second, so data will be permanently lost // There will be fewer samples per second, so the original sound cannot be re-created	1
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QUESTION

- (ii) Identify whether the lossy or lossless compression method is more appropriate for real-time bit streaming. Justify your answer.

Compression method

Justification

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[3]

MARK SCHEME

2(b)(ii)	No mark for choice but lossy is most appropriate 1 mark each to max 3 e.g. • Reduces file size more than lossless • so significantly less bandwidth / data is needed • so buffering is reduced even more than with lossless • Data can be removed which cannot be seen • reducing quality without impacting experience • for example, because resolution of video can be reduced // sample rate of audio can be reduced	3
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QUESTION

- (iii) A bitmap image can be compressed using lossy compression.

Explain the reasons why lossy compression is often suitable for a bitmap image.

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MARK SCHEME

6(b)(iii)	1 mark for each bullet point (max 2) e.g. • The change may not be noticeable // Data removed is usually not noticed by the human eye • ... for example, changes in shade/detail • It produces a larger decrease in file size compared to lossless // Lossy decreases file size considerably	2
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9618_w24_qp_13_sc_6.c.ii

QUESTION

(ii) Describe **one** method of compressing a sound file using lossy compression.

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..... [2]

MARK SCHEME

6(c)(ii)	1 mark for each correct point and 1 mark for matching expansion e.g. • Decrease sample rate • ... fewer samples/readings/measurements stored per second // fewer bits per second stored • Decrease sample resolution • ... fewer bits per sample/reading/measurement // each sample has fewer bits • Sound outside of set/human hearing range is removed • ... fewer measurements are stored / decreases the number of possible binary values so fewer bits are stored	2
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9618_w24_qp_12_sc_7.a.iii

QUESTION

(iii) Identify **one** lossless method of compressing an image.

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MARK SCHEME

7(a)(iii)	1 mark for correct method For example: Run-Length Encoding	1
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9618_s24_qp_13_sc_2.b.i

QUESTION

(b) The photograph is compressed before being uploaded to a web server.

(i) Give **three** benefits of this photograph being compressed using lossy compression instead of lossless compression.

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[3]

MARK SCHEME

2(b)(i)	1 mark each to max 3 : - The file takes less storage space on the web server than if lossless compression was used - The file is faster to upload/download to/from the server than if lossless compression was used - The file uses less bandwidth to transmit than if lossless compression was used - The file consumes less data allowance than if lossless compression was used	3
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QUESTION

- 6 (a) A real-time video of a music concert needs to be streamed to subscribers.

Tick (✓) **one** box to identify the most appropriate type of compression **and** justify your answer.

Lossy	Lossless
☐	☐

Justification

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..... [3]

MARK SCHEME

6(a)	1 mark for each bullet point (max 3) Lossy compression (ticked) • Loss of quality will not be noticed • Needs to be viewed in real time so less bandwidth needed if file size smaller • Smaller file sizes will reduce buffering so the video will play more smoothly • Viewers may watch on different devices, so may not need high quality resolution Lossless compression (ticked) • Original recording may not have been made in high resolution • Could be streaming to high bandwidth devices • The reduction in the file size is sufficient for the receiving device • Viewers do not want any loss of quality	3
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QUESTION

- (ii) RLE is an example of lossless compression.

Explain why lossless compression is more appropriate than lossy compression for a text file.

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MARK SCHEME

8(c)(ii)	1 mark for each bullet point: - all the data is required // no data can be lost ... otherwise text file will be corrupted / not make sense	2
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QUESTION

- (ii) Riya needs to email the photograph. She compresses the photograph before sending it using an email.

Describe **two** lossy methods that Riya can use to compress the image.

Method 1

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Method 2

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..... [4]

MARK SCHEME

5(b)(ii)	1 mark for each bullet point to max 2 for each method • Reduce bit depth • ... reduces the number of bits per colour / pixel which means each pixel has fewer bits • Reduce colour palette // reduce number of colours • ... fewer colours mean fewer bits needed to store each colour • Reduce image resolution • ... fewer pixels per unit measurement means less binary to store	4
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9618_w21_qp_11_sc_7.b.ii

QUESTION

(ii) Bobby uses lossless compression.

Describe how lossless compression can compress the sound file.

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MARK SCHEME

7(b)(ii)	1 mark per bullet point to max 2 e.g. • Reduce amplitude to only the range used • ... limited amplitudes mean fewer bits per sample • Run-length-encoding • ... Where consecutive sounds are the same record the binary value of the sound and number of times it repeats • Record the changes instead of the actual sounds	2
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QUESTION

(b) The image is compressed using lossless compression.

Identify **one** method of lossless compression that can be used to compress the image **and** describe how the method will reduce the file size.

Lossless compression method

Description

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[3]

MARK SCHEME

1(b)	1 mark for naming method, 1 mark per description to max 2 • Run-length encoding • Replace sequences of the same colour pixel • ... with colour code and number of identical pixels	3
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Show understanding of how a text file, bitmap image, vector graphic and sound file can be compressed

9618_w25_qp_12_sc_6.a

QUESTION

- 6 (a) A sound file is compressed by reducing the sampling rate.

State whether this is lossless or lossy compression. Justify your choice.

Type of compression

Justification

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[1]

MARK SCHEME

6(a)	1 mark for correct answer Type of compression: lossy Justification: There will be fewer samples per second, so data will be permanently lost // There will be fewer samples per second, so the original sound cannot be re-created	1
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9618_w25_qp_11_sc_7.d.iii

QUESTION

- (iii) Explain how a bitmap image is compressed using run-length encoding (RLE).

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MARK SCHEME

7(d)(iii)	1 mark per bullet point, max 2 marks - Sequences of consecutive identical colours / pixels ... are stored as the colour value and the number of times it occurs consecutively	2
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9618_w24_qp_13_sc_6.c.ii

QUESTION

(ii) Describe **one** method of compressing a sound file using lossy compression.

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MARK SCHEME

6(c)(ii)	1 mark for each correct point and 1 mark for matching expansion e.g. - Decrease sample rate ... fewer samples/readings/measurements stored per second // fewer bits per second stored - Decrease sample resolution ... fewer bits per sample/reading/measurement // each sample has fewer bits - Sound outside of set/human hearing range is removed ... fewer measurements are stored / decreases the number of possible binary values so fewer bits are stored	2
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9618_w24_qp_12_sc_7.a.iii

QUESTION

(iii) Identify **one** lossless method of compressing an image.

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MARK SCHEME

7(a)(iii)	1 mark for correct method For example: • Run-Length Encoding	1
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9618_w24_qp_11_sc_4.b.ii

QUESTION

(ii) Describe **one** lossless method of compressing a text file.

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MARK SCHEME

4(b)(ii)	1 mark for each bullet point (max 3) • Run-Length Encoding // RLE • Repeated sequences of the same characters are replaced by • ... a single copy of the character • ... and a count of the number of characters	3
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9618_s24_qp_13_sc_2.b.ii

QUESTION

(ii) Explain how run-length encoding (RLE) will compress the photograph.

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MARK SCHEME

2(b)(ii)	1 mark each: - Identifies consecutive repeating pixels of the same colour - Stores the colour /pattern and the number of times it repeats	2
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9618_s23_qp_11_sc_1.c.ii

QUESTION

(ii) An image can be compressed using run-length encoding (RLE).

Explain the reasons why RLE may **not** reduce the file size of a bitmap image.
Give **one** example in your answer.

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MARK SCHEME

1(c)(ii)	1 mark each to max 2 for explanation; 1 mark for an image related example - RLE stores a colour and the number of times it occurs consecutively - An image may not have many sequences of the same colour - It would need to store each colour and then the count/number 1 which adds data Example: - Red-Green-Blue would become Red 1 Green 1 Blue 1	3
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QUESTION

- (c) A second bitmap image is stored using a colour depth of 8 bits per pixel.
- The file is compressed using run-length encoding (RLE).
- (i) The table shows the compressed and uncompressed values for parts of the image file.
- Each colour of the pixel in the image is represented by a hexadecimal value.
- Complete the table. The first row has been completed for you.

Uncompressed image	RLE compressed image
EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA
	2AB 2FF 11D 167
32 32 80 81 81	

[2]

MARK SCHEME

8(c)(i)	1 mark for each correct underlined part:	2								
<table><tr><th>Uncompressed sound</th><th>RLE compressed sound</th></tr><tr><td>EA F1 F1 F2 F2 F2 EA</td><td>1EA 2F1 3F2 1EA</td></tr><tr><td><u>AB AB FF FF 1D 67</u></td><td>2AB 2FF 11D 167</td></tr><tr><td>32 32 80 81 81</td><td><u>232 180 281</u></td></tr></table>			Uncompressed sound	RLE compressed sound	EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA	<u>AB AB FF FF 1D 67</u>	2AB 2FF 11D 167	32 32 80 81 81	<u>232 180 281</u>
Uncompressed sound	RLE compressed sound									
EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA									
<u>AB AB FF FF 1D 67</u>	2AB 2FF 11D 167									
32 32 80 81 81	<u>232 180 281</u>									

QUESTION

- (c) An Operating System may include a utility program to compress text files.
- Describe **one** appropriate method of compressing a text file.
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- [3]

MARK SCHEME

7(c)	1 mark for each bullet point (max 3): • lossless compression • Run Length Encoding • repeated sequences of <u>characters</u> are replaced by • ... a single copy of the character • ... and a counter of the number of characters	3
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9618_w21_qp_12_sc_5.b.ii

QUESTION

- (ii) Riya needs to email the photograph. She compresses the photograph before sending it using an email.

Describe **two** lossy methods that Riya can use to compress the image.

Method 1

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Method 2

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[4]

MARK SCHEME

5(b)(ii)	1 mark for each bullet point to max 2 for each method • Reduce bit depth • ... reduces the number of bits per colour / pixel which means each pixel has fewer bits • Reduce colour palette // reduce number of colours • ... fewer colours mean fewer bits needed to store each colour • Reduce image resolution • ... fewer pixels per unit measurement means less binary to store	4
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QUESTION

(ii) Bobby uses lossless compression.

Describe how lossless compression can compress the sound file.

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..... [2]

MARK SCHEME

7(b)(ii)	1 mark per bullet point to max 2 e.g. • Reduce amplitude to only the range used • ... limited amplitudes mean fewer bits per sample • Run-length-encoding • ... Where consecutive sounds are the same record the binary value of the sound and number of times it repeats • Record the changes instead of the actual sounds	2
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QUESTION

(b) The image is compressed using lossless compression.

Identify **one** method of lossless compression that can be used to compress the image **and** describe how the method will reduce the file size.

Lossless compression method

Description

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[3]

MARK SCHEME

1(b)	1 mark for naming method, 1 mark per description to max 2 • Run-length encoding • Replace sequences of the same colour pixel • ... with colour code and number of identical pixels	3
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